

Executive Functioning and Imperative Programming

Executive Functioning

1. What are the three major executive functions?

- Working Memory
- Cognitive Flexibility
- Inhibitory Control

2. What are the twelve executive functions from the self assessment we took?

- Response Inhibition
- Working Memory
- Emotional Control
- Sustained Attention
- Task Initiation
- Planning/Prioritization
- Organization
- Time Management
- Goal-directed persistence
- Flexibility
- Metacognition
- Stress Tolerance

3. Please identify one or two of the executive functions that you'd like to work on.

I would like to work on task initiation the most this semester as I scored low on that executive function and I believe it causes many problems for me.

I plan to employ 3 interventions this semester to help me get better at initiating tasks. First and I think the most important thing is having a routine. Years ago, no matter what, at 8PM I would do some kind of academic work. This would often lead me to be ahead in many classes and I always had the advantage of knowing what is happening in class. I will go back to a 7 PM routine where at that time I will work on tasks; this includes studying or even practicing certain concepts if I shouldn't work on a certain assignment or currently don't have a certain assignment for some reason.

Another Intervention I will use will be making a checklist to help me avoid distractions. Oftentimes, I feel like if I don't really understand why I am doing a certain task, I'm not as interested in doing it. I think before starting a task, if I try to understand why I am doing a task, what I need to do it, and how do I start, it will help me mentally engage with

the work. I also think if I can fall into this thought process early into an assignment, I can talk to professors and ask specific questions that can help me stay further engaged. Finally the third intervention I plan to use is doing some kind of warm up. This is something I used to do a lot years ago, and still somewhat do to this day. I start with some kind of warm up assignment or exercise with the task I am trying to complete. This helps wake up my brain and makes me want to do harder tasks which is something that can help me stay motivated while doing difficult work.

<https://ctl.stanford.edu/students/getting-started-tasks>
<https://www.youtube.com/watch?v=sQv26WyiPCs>

Imperative Programming

1. Definition of Paradigm outside of computer programming context.

A fundamental framework that shapes a perspective or approach to a particular subject or idea.

2. What is our definition of a paradigm in programming languages?

A fundamental style or approach to solving problems using a programming language based on their features.

3. What does the word “Imperative” Mean?

Absolutely necessary or required to happen. It is a direct command.

4. What three things are required for imperative programming?

Assignment: ability to set and change the value of variables in memory

Sequencing: the order instructions are executed

Control Flow: the order a programs statements and instructions are executed using loops, conditionals, or functions.

5. What three things do we use to “Program” a Turing machine?

Tape Alphabet: The symbols that can be read from or written to the tape

Set of States: The different modes or states a machine can be in

Transition Function: The rules that tell the machine what symbol to write, which way to move the head, and what state to enter next.

6. What does it mean for a language to be “Turing Complete”?

A language is Turing Complete if you can use it to compute the same things a universal turing machine can compute given enough time and resources.

7. Find two examples of Turing Complete Languages that are imperative.

R:

```
factorial <- function(n) {  
  if(n <= 1) {  
    return(1)  
  } else {  
    return(n * recur_factorial(n-1))  
  }  
}
```

JavaScript:

```
function factorial(n) {  
  if (n <= 1) {  
    return 1;  
  } else {  
    return n * factorial(n - 1);  
  }  
}
```

8. Find two examples of Programming Languages that aren't Turing Complete.

JSON: Text-based data format that creates objects that can be stored or transmitted. Primarily used to exchange data between servers and clients, but has many other uses, such as for dictionaries or configuration files. The reason it is not Turing-complete is that it doesn't contain any logic; therefore, it cannot perform any computations, similar to CSV.

CSS: A style sheet language used to describe the visual appearance of documents written in markup languages. It is used to define the design, such as the layout, color, fonts, or displays for different devices. Similar to JSON, however, it lacks the control flow required for it to perform any computations; therefore, it is not Turing-complete.